display. The device features an RGB camera, depth sensor, multi-array microphone, and a custom processor running proprietary software, which provides full-body 3D motion capture, facial recognition, and voice recognition capabilities.

The Project Natal sensor's microphone array enables the Xbox 360 to conduct acoustic source localisation and ambient noise suppression, allowing for things such as headset-free party chat over Xbox Live.

The depth sensor consists of an infrared projector combined with a monochrome CMOS sensor, and allows the Project Natal sensor to see in 3D under any ambient light conditions. The sensing range of the depth sensor is adjustable, with the Project Natal software capable of automatically calibrating the sensor based on gameplay and the player's physical environment, such as the presence of couches.

At E3 2009, the MS team demonstrated a skeletal mapping technology that is capable of simultaneously tracking up to four users for motion analysis, with a feature extraction of 48 points of interest on a human body at a frame rate of 30 hertz.

In fact, depending on the person's distance from the sensor, Project Natal is capable of tracking models that can identify individual fingers!

Although no official release date has been set for the technology, Microsoft is expected to release it sometime in 2010.

4.2 Stereoscopic 3D Games

While video game technologies are rapidly progressing with new graphics cards and faster processors, the output module has remained the same – a television set. However, it is not as if the TV industry has remained placid; we have seen the conversion from CRT-based bulky sets to thin LCD screens, and the jump from standard 4:3 video to HD and full HD resolutions.

The next jump being predicted is the jump to stereoscopic 3D television, possibly without the use of those red-and-blue 3D glasses. Whether the glasses will stay or go is a discussion for another forum, but what is certain is that making standard TVs render stereoscopic 3D images is definitely on the anvil. And the gaming industry would be the ideal platform to capitalise.

Imagine a combination of Project Natal and such a stereoscopic 3D TV, where you act out a boxing

Whether the red-and-blue glasses will stay or go is irrelevant; what is certain is that making standard TVs render stereoscopic 3D images is definitely on the anvil, and the gaming industry would be the ideal platform to capitalize on this